

Chapter 1: The Drug Discovery Pipeline & AI Revolution

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1.1 The Traditional Drug Discovery Pipeline

Bringing a new medicine to market is a multi-stage filtering process. Each stage exists to kill bad candidates as cheaply as possible, before they reach a stage where failure is expensive. The stages, in order, are broadly:

1. **Target identification & validation** — establishing that modulating a specific biomolecule (usually a protein) plausibly changes the course of a disease, using genetic, biochemical, or clinical evidence.
2. **Hit identification** — screening large compound libraries (physically, via high-throughput screening, or virtually, via computational docking/similarity search) to find molecules that bind the target.
3. **Hit-to-lead & lead optimization** — iteratively modifying hit compounds to improve potency, selectivity, and drug-like properties (solubility, metabolic stability, low toxicity) while tracking structure–activity relationships (SAR).
4. **Preclinical development** — in vitro and animal studies of efficacy, pharmacokinetics (absorption, distribution, metabolism, excretion — ADME), and toxicology, required before human dosing.
5. **Clinical trials** — Phase I (safety, small healthy/patient cohorts), Phase II (efficacy signal, dose-finding), and Phase III (large, randomized, confirmatory efficacy and safety) trials in human subjects.
6. **Regulatory review and approval** — submission of the accumulated evidence (e.g., a New Drug Application to the US FDA) for marketing authorization.

A candidate can fail at any stage, and the cost of failure grows by orders of magnitude the later it happens — a compound that fails in early screening costs a lab reagents and a few days; a compound that fails in Phase III has consumed years and hundreds of millions of dollars enrolling and monitoring patients. This asymmetry is the central economic argument for pushing predictive power as early into the pipeline as possible, which is exactly where the computational and AI methods covered in this book are concentrated.

The economics of attrition: Eroom’s Law

Despite decades of technological improvement — combinatorial chemistry, high-throughput screening, genomics, and now machine learning — the *inflation-adjusted* number of new drugs approved per billion US dollars of R&D spending has been in long-term decline. Scannell et al. (2012) documented that this figure roughly halved every nine years from 1950 to the early 2010s, an approximately 80-fold fall in R&D efficiency over that period, in direct contrast to the exponential improvement in computing power described by Moore’s Law. They named this trend **Eroom’s Law** (“Moore’s Law” spelled backwards) to emphasize the contrast. If $N(t)$ is the number of new drugs approved per billion inflation-adjusted dollars of R&D spend at year t , the observed trend is approximately:

$$N(t) \approx N_0 \cdot 2^{-\frac{t-t_0}{T}}, \quad T \approx 9 \text{ years}$$

where T is the empirical halving period. Scannell et al. attribute this decline to four compounding factors rather than a single cause: the “better than the Beatles” problem (new drugs must beat an ever-improving standard of care, not merely a placebo), increasingly risk-averse regulatory and reimbursement environments, the tendency to increase R&D spending as a response to past failure rather than as a cause of future success, and a “basic research–brute force” bias toward molecular-target reductionism at the expense of systems-level understanding of disease.

The cost and risk of a single approved drug

Two numbers define the economics of drug development: how much it costs, and how likely a candidate is to survive the pipeline at all.

Wouters, McKee, and Luyten (2020) estimated development costs using publicly disclosed data (SEC filings, FDA approval records, and ClinicalTrials.gov) for 63 of the 355 new therapeutic agents the FDA approved between 2009 and 2018. After capitalizing costs at a real cost of capital of 10.5% per year to account for the time value of money and the cost of capital tied up in failed programs, they estimated a **median** R&D investment of \$985.3 million (95% CI: \$683.6M–\$1228.9M) and a **mean** of \$1,335.9 million (95% CI: \$1,042.5M–\$1,637.5M) per approved drug, in 2018 US dollars — figures that varied enormously by therapeutic area, from a median of \$765.9M for nervous-system drugs to \$2,771.6M for oncology and immunomodulatory drugs. (A 2022 erratum to this paper corrected errors found in the underlying source data; the corrected figures above are the ones reported in that correction. See References.)

The reason these figures are so large is that they are *risk-adjusted*: the direct, out-of-pocket cost of running one clinical trial is much smaller than the effective cost of getting one drug approved, because most candidates that enter clinical trials never make it out. A standard way to express this is to capitalize the cost of each phase forward to the launch date and divide by the cumulative probability of a candidate that has reached the start of that phase eventually being approved:

$$E[\text{Cost}_{\text{launch}}] = \sum_{i=1}^n \frac{C_i}{\text{POS}_i} \cdot (1+r)^{T_{\text{launch}}-t_i}$$

where C_i is the direct cost of phase i , POS_i is the probability that a candidate entering phase i is ultimately approved, r is the real cost of capital, and t_i is the phase’s timing relative to launch at T_{launch} . This is the general shape of the risk-adjustment framework used throughout the drug-development-cost literature (it is a simplified pedagogical form, not a specific paper’s exact computation); the key takeaway is that POS_i — the attrition rate — dominates the equation, because it appears in the denominator of every term.

Attrition itself is substantial and disease-dependent. Using a sample of over 406,000 clinical trial records for more than 21,000 compounds run between 2000 and 2015, Wong, Siah, and Lo (2019) found that oncology drugs had only a 3.4% end-to-end clinical success rate in their sample — notably lower than the 5.1% figure widely cited from earlier, smaller industry-curated datasets — though this rate has been highly variable year to year (it fell to 1.7% in 2012 before rising to

8.3% in 2015). Hay et al. (2014) independently surveyed clinical success rates across the broader industry using one of the largest datasets assembled at the time. Both studies agree on the qualitative picture even where the exact percentages differ: **the overwhelming majority of clinical candidates fail**, and failure is concentrated in efficacy (the drug doesn’t work well enough) far more than in safety.

This is the economic backdrop against which every method in this book should be judged. A computational technique earns its place in the pipeline only if it plausibly increases POS_i at some stage, or reduces C_i , or both — not because it is fashionable.

1.2 The AI Paradigm Shift

Classical computer-aided drug design relies on hand-engineered descriptors and rule-based heuristics (Chapter 2 uses one of the oldest and still most useful examples, Lipinski’s Rule of Five). Over the last decade, machine learning has been layered on top of — and in some cases has begun to replace — this classical toolkit. Vamathevan et al. (2019), in a widely cited review, characterize opportunities for ML across every stage of the pipeline described in §1.1: target identification and validation, prognostic biomarker discovery, and the analysis of high-dimensional data such as digital pathology images in clinical trials. Their central caveat is as important as their central claim: ML performs best on “well-specified questions with abundant, high-quality data,” and its adoption is currently limited more by interpretability and repeatability of results than by raw predictive accuracy.

It is useful to organize the computational methods used throughout this book into three broad, overlapping families:

- **Predictive models** answer a question about a specific molecule or protein: will this compound bind this target (Chapters 5–6)? Is this molecule likely to be toxic (Chapter 5)? What is this protein’s 3D structure (Chapter 9)? These are typically supervised learning problems — classification or regression on a labeled dataset — and their scientific value is bounded by the quality and size of that dataset, which is precisely why chemical and structural databases like ChEMBL and the PDB (introduced in §1.4) are foundational infrastructure for the entire field.
- **Generative models** propose new molecules or sequences rather than scoring existing ones: variational autoencoders and diffusion models over molecular graphs or 3D coordinates (Chapter 7), autoregressive transformers over SMILES/SELFIES strings (Chapter 7), and diffusion models over protein backbones (Chapter 10). These models shift the bottleneck in early discovery from *screening* a fixed library to *designing* candidates directly in the region of chemical or sequence space likely to satisfy a set of constraints.
- **Physics-informed models** incorporate known physical or chemical constraints — conservation laws, symmetries, or explicit energy functions — directly into the model architecture or training objective, rather than relying on a purely data-driven fit. Equivariant graph neural networks that respect 3D rotational and translational symmetry (used in Chapter 6 and Chapter 9) and neural network potentials that approximate quantum-mechanical energies at classical-simulation speed (Chapter 12) are both examples: the model is constrained to only ever produce

physically consistent outputs, which improves data efficiency and generalization compared with an unconstrained model trained on the same data.

The clearest illustration of what this paradigm shift can achieve is structure prediction. Determining a protein’s 3D structure experimentally (by X-ray crystallography, NMR, or cryo-EM) can take months to years per structure and does not always succeed. Jumper et al. (2021) showed that AlphaFold, a deep learning system, could predict protein structures from amino acid sequence alone at accuracy competitive with experimental methods for a large fraction of targets in the CASP14 blind assessment — work that has since made high-confidence structural models available for essentially the entire sequenced proteome, at a computational cost orders of magnitude below experimental determination. Chapter 9 covers AlphaFold and related methods (ESMFold, RoseTTAFold) in depth; it is introduced here because it is the single clearest data point for the claim that this book is built on: that the AI paradigm shift is not merely accelerating the classical pipeline, it is changing which questions are computationally tractable at all.

None of this makes the classical pipeline in §1.1 obsolete. Every predictive, generative, and physics-informed model in this book is trained on data that ultimately comes from wet-lab experiments — assays run against real targets, structures solved by real crystallographers, outcomes from real clinical trials. AI does not remove the pipeline; it changes where effort within the pipeline is spent, ideally shifting attrition as early and as cheaply as possible, per the economic argument in §1.1.

1.3 Modalities Overview: Small Molecules vs. Macromolecules/Biologics

Everything described so far applies differently depending on what kind of molecule is being designed. The book covers two broad modalities, which differ enough in their representation, design constraints, and manufacturing that they are treated with largely separate computational toolkits in Parts I–II (small molecules) and Part III (macromolecules).

Small molecules are low-molecular-weight organic compounds, synthesized chemically rather than produced biologically, and — for orally dosed drugs — usually expected to satisfy heuristics like Lipinski’s Rule of Five: no more than 5 hydrogen-bond donors, no more than 10 hydrogen-bond acceptors, a molecular weight under 500 Da, and a calculated LogP under 5, on the premise that violating more than one of these substantially reduces the probability of good oral absorption and permeability (Lipinski et al., 2001). Chapter 2’s hands-on project computes exactly these four properties with RDKit. Small molecules are represented computationally as 2D graphs, 1D strings (SMILES, SELFIES, InChI — Chapter 2), fingerprints (Chapter 2), or 3D conformers (Chapter 2), and their chemical space is explored using the fingerprint similarity, generative, and docking methods in Chapters 2, 7, and 11.

Macromolecules / biologics — therapeutic proteins, peptides, and antibodies — are produced biologically (typically via recombinant expression in engineered cells) rather than synthesized chemically, are orders of magnitude larger (a typical IgG monoclonal antibody is roughly 150 kDa, versus a few hundred Da for a small molecule), and cannot generally be dosed orally because they are degraded by the digestive system, so they are usually delivered by injection or infusion. Their size and complexity give them properties small molecules structurally cannot match — very high

target specificity and affinity, engineerable multi-domain architectures (e.g., bispecific antibodies) — at the cost of manufacturing complexity, immunogenicity risk, and a much larger, higher-dimensional design space (a single antibody’s antigen-binding region alone involves optimizing complementarity-determining region loops built from 20 possible amino acids at each position). Part III of this book (Chapters 8–10) covers the representations (sequences, contact maps, 3D structural graphs) and generative methods (inverse folding, backbone diffusion) specific to this modality.

The choice of modality for a given disease target is itself a design decision with no universally correct answer: small molecules are cheaper to manufacture and can reach intracellular targets that biologics generally cannot, while biologics can achieve selectivity against targets (such as flat protein–protein interaction surfaces) that are notoriously difficult to drug with small molecules. Recognizing which modality is appropriate for a given target — and which computational tool-chain in this book therefore applies — is one of the first judgment calls in any real drug discovery project, which is why this book treats it as the subject of its very first hands-on exercise, below.

1.4 Hands-on Project: Retrieving Real Chemical & Structural Data

The rest of this book works with data pulled from two public infrastructure databases that anchor the whole field:

- **ChEMBL** (Mendez et al., 2019), maintained by EMBL-EBI, is a curated, open-access database of bioactivity data — which compounds were tested against which biological targets, and what activity (e.g., IC₅₀, K_i) was measured. As of release ChEMBL_37 (May 2026), it holds bioactivity data for roughly 2.9 million distinct compounds against over 18,000 targets, drawn from more than 100,000 publications.
- **The Protein Data Bank (PDB)** (wwPDB consortium, 2019) is the single global archive of experimentally determined 3D structures of proteins, nucleic acids, and their complexes. As of August 2026 it holds 258,403 released structures.

The hands-on project for this chapter does not analyze this data yet — that begins in Chapter 2 (chemical similarity) and Chapter 9 (structure prediction) — it establishes the environment and the *programmatic* retrieval pattern that the rest of the book builds on: querying both databases’ public REST APIs directly from Python, rather than manually downloading files through a web browser. This is a deliberate choice: a reproducible computational pipeline should be able to re-fetch its own inputs from source.

The project code lives in this chapter’s folder (`ch01_drug_discovery_pipeline/`) and targets a single, concrete example throughout: **EGFR** (epidermal growth factor receptor, ChEMBL target CHEMBL203), a receptor tyrosine kinase that is one of the most extensively studied oncology drug targets and the target used in this book’s Chapter 16 capstone project. It retrieves:

1. **Target metadata and bioactivity data from ChEMBL** — confirming the target identity (preferred name, organism, target type) and pulling a page of measured bioactivities (assay type, measured value, units, and the tested compound’s SMILES string) for EGFR.

2. **Structural metadata and a 3D structure file from the PDB** — for PDB entry 1M17, the crystal structure of the EGFR tyrosine kinase domain bound to the inhibitor erlotinib, retrieving both the entry’s metadata (title, resolution, release date) and the structure file itself.

See README.md in this folder for setup and usage instructions, and `fetch_data.py` for the implementation. The accompanying test suite (`tests/test_fetch_data.py`) exercises the same functions against the live APIs, so running the tests is itself a reproducibility check on the retrieval pipeline.

A note on Google Colab

Every dependency this project needs (requests, part of a standard Python installation’s ecosystem) is preinstalled on Google Colab’s default runtime, so the entire project runs unmodified in a fresh Colab notebook — install nothing, add a cell with `!git clone` or paste `fetch_data.py`’s contents, and run. No GPU or special runtime is required for this chapter’s project; later chapters that require a GPU runtime will say so explicitly.

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Live data cited in §1.4 (ChEMBL release version and count, PDB entry count) were retrieved directly from the ChEMBL and RCSB PDB REST APIs on 2026-08-16; see `fetch_data.py` to reproduce.